Formulas for 47202 Introduction to Future Energy

Version 16-10-2026

Contents

Constants 2
Metric prefixes 3
SI units 3
The base units in the SI system 3
Additional units in the SI system 3
Definitions 4
General 4
Thermodynamic systems 4
Basic physics 5
Mechanical energy 5
Moments of inertia, *I* 5
Thermal energy 6
Electrical energy 6
Gasses 6
Thermodynamics 7
Electrochemistry 8
Solar Energy 9
Wind Energy 10
Bioenergy 12
Nuclear physics 12
Economics 14

Constants

Constant	Value
Speed of light, c	$2.998 \cdot 10^{8}$ m s^{-1}
Avogadro's constant (Avogadro's number), N_A	$6.022 \cdot 10^{23}$ mol^{-1}
Elementary charge (of electron or proton), e	$1.602 \cdot 10^{-19}$ C
Electronvolt, eV	$1.602 \cdot 10^{-19}$ J
Atomic mass unit, amu	$1.6605 \cdot 10^{-27}$ kg
Mass of electron, m_e (from NIST)	$9.109 \cdot 10^{-31}$ kg $5.4858 \cdot 10^{-4}$ amu
Mass of proton, m_p (from NIST)	$1.6726 \cdot 10^{-27}$ kg 1.007276 amu
Mass of neutron, m_n (from NIST)	$1.6749 \cdot 10^{-27}$ kg 1.008665 amu
Mass of Hydrogen atom (protium, $^{1}_{1}\mathrm{H}$), m_H	$1.6735 \cdot 10^{-27}$ kg 1.007825 amu
Faraday's constant, F	96 485 C mol^{-1}
Universal gas constant, R	8.314 J mol^{-1}K^{-1} $8.314 \cdot 10^{-2}$ L bar mol^{-1}K^{-1} 0.0821 L atm mol^{-1} K^{-1}
Vacuum permittivity, $\mathcal{E}$	$8.854 \cdot 10^{-12}$ C^2 J^{-1} m^{-1} $8.854 \cdot 10^{-12}$ F m^{-1}
Gravitational constant, G	$6.6743 \cdot 10^{-11}$ m^3 kg^{-1} s^{-2} $6.6743 \cdot 10^{-11}$ N m^2 kg^{-2}
Acceleration of gravity (value slightly varying), g	9.81 m s^{-1}

Metric prefixes

Name	Symbol	Base	Name	Symbol	Base
kilo	k	10^{3}	milli	m	10^{-3}
mega	M	10^{6}	micro	μ	10^{-6}
giga	G	10^{9}	nano	n	10^{-9}
tera	T	10^{12}	pico	p	10^{-12}
peta	P	10^{15}	femto	f	10^{-15}
exa	E	10^{18}	atto	a	10^{-18}
zetta	Z	10^{21}	zepto	z	10^{-21}

SI units

The base units in the SI system

Quantity	Name (SI unit)	Symbol
Time	second	s
Length	metre	m
Mass	kilogram	kg
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Additional units in the SI system

Quantity	Name (SI unit)	Unit short	Intuitively	From base units
Energy	joule	J	N·m	$kg{\cdot}m^{2}{\cdot}s^{-2}$
Power	watt	W	V·A	$kg{\cdot}m^{2}{\cdot}s^{-3}$
Force	newton	N		$kg{\cdot}m{\cdot}s^{-2}$
Pressure	pascal	Pa	$N{\cdot}m^{-2}$	$kg{\cdot}m^{-1}{\cdot}s^{-2}$
Electric potential	volt	V	$J{\cdot}C^{-1}$	$Kg{\cdot}m^{2}{\cdot}s^{-3}{\cdot}A^{-1}$
Charge	coulomb	C		s·A
Capacitance	farad	F	$C{\cdot}V^{-1}$	$kg^{-1}{\cdot}m^{-2}{\cdot}s^{4}{\cdot}A^{2}$
Resistance	ohm	Ω	$V{\cdot}A^{-1}$	$kg{\cdot}m^{2}{\cdot}s^{-3}{\cdot}A^{-2}$
Conductivity	siemens	S	Ω^{-1}	$kg^{-1}{\cdot}m^{-2}{\cdot}s^{3}{\cdot}A^{2}$

Definitions

General

Name	Definition
Capacity factor	$CF = \frac{\text{Actual power}}{\text{Rated power}}$ $CF = \frac{\text{Actual energy}}{\text{Rated energy}}$ $CF = \frac{\text{Actual annual energy}}{\text{Rated annual energy}}$
Efficiency	$\eta = \frac{\text{What you get}}{\text{What you pay}}$
Standard condition	Exact: all species at unit activity Approximated: all gases at 1 bar, all solutions at 1 molar. Often assumed, T = 298.15 K
Pure condensed phased (pure liquids, pure solids)	Activity, a ≡ 1
Higher heating value, HHV	Enthalpy of combustion as a positive value with product water as a *liquid*
Lower heating value, LHV	Enthalpy of combustion as a positive value with product water as a *gas*

Thermodynamic systems

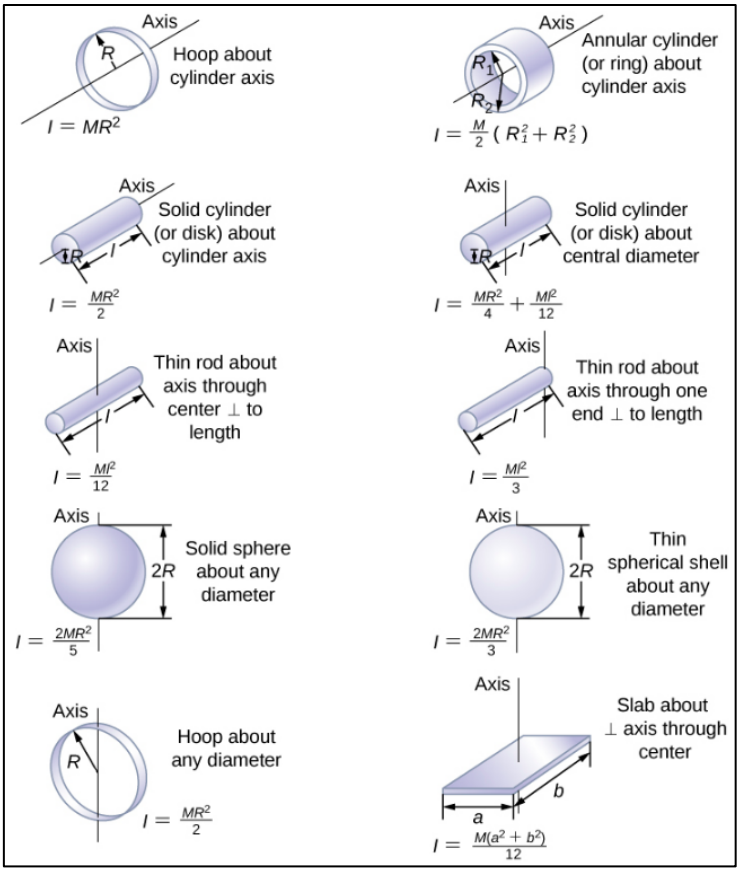
Description	Definition
An *isolated* system	A system that does not exchange heat, work or matter with the surroundings.
A *closed* system	A system that does not exchange matter with the surroundings.
An *isothermal* process	A process carried out at constant temperature.
An *isobaric* process	A process carried out at constant pressure.
An *isochoric* process	A process carried out at constant volume.
An *exothermic* (or *exothermal*) process	A process that evolves thermal energy.
An *endothermic* (or *endothermal*) process	A process that consumes thermal energy.

Basic physics

Mechanical energy

Name	Formula
Work	$W = Fx$
Power (P) and energy (E)	$P = \frac{E}{t}$
Force (F) and acceleration (a) (linear)	$F = ma$
Torque (τ) and angular acceleration (α)(rotational)	$\tau = I\alpha$
Kinetic energy (linear)	$E_{\text{kin}} = ½mv^2$
Kinetic energy (rotational)	$E_{\text{kin}} = ½I\omega^2$
Potential energy	$E_{\text{pot}} = mgh$
Gravitational force	$F = G\frac{m_1 m_2}{r^2}$

Moments of inertia, *I*



Thermal energy

Name	Formula
Sensible heat (constant heat capacity)	$q = mc\Delta T$
Carnot's law	$\eta_{\text{carnot}} = \frac{T_{\text{h}} - T_{\text{c}}}{T_{\text{h}}}$ $\eta_{\text{carnot}} = 1 - \frac{T_{\text{c}}}{T_{\text{h}}}$

Electrical energy

Name	Formula
Power and current	$P = I\Delta U$ $P = RI^2$
Voltage drop	$\Delta U = RI$
Energy and power	$E_{\text{el}} = Pt$
Charge and current	$Q = It$
Ohm's law	$U = RI$
Resistors in series	$R_{\text{tot}} = R_1 + R_2 + R_3 \ \ldots$
Resistors in parallel	$\frac{1}{R_{\text{tot}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ \ldots$
Electrical force	$F = k_e \frac{q_1 q_2}{r^2}$
Energy and charge	$\Delta E_{\text{el}} = q\Delta U$
Capacitance	$Q = UC$
Capacitance	$C = \varepsilon_{\text{r}} \varepsilon_0 \frac{A}{d}$
Capacitors in series	$\frac{1}{C_{\text{tot}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \ \ldots$
Capacitors in parallel	$C_{\text{tot}} = C_1 + C_2 + C_3 \ \ldots$
Energy of charged capacitor	$W_{\text{e}} = ½CU^2$

Gasses

Name	Formula
Ideal gas (perfect gas)	$pV = nRT$

Thermodynamics

Name	Formula
Enthalpy	$H \equiv U + pV$
Entropy	$\mathrm{d}S \geq \frac{\mathrm{d}Q}{T}$ $\Delta S = \frac{Q_{\mathrm{rev,min}}}{T}$
Entropy of states	$S_{\mathrm{gas}} >>> S_{\mathrm{liquid}} > S_{\mathrm{solid}}$
Gibbs free energy	$\Delta G_{\mathrm{r}} = \Delta H_{\mathrm{r}} - T\Delta S_{\mathrm{r}}$ $\Delta G = \Delta H - T\Delta S$
Carnot's law	$\eta_{\mathrm{carnot}} = \frac{T_{\mathrm{h}} - T_{\mathrm{c}}}{T_{\mathrm{h}}}$ $\eta_{\mathrm{carnot}} = 1 - \frac{T_{\mathrm{c}}}{T_{\mathrm{h}}}$
Maximum work from heat	$W_{\mathrm{max}} = \eta_{\mathrm{carnot}} Q$
Coefficient of performance (COP) (Heat pump)	$\mathrm{COP} = \frac{\dot{Q}_{\mathrm{h}}}{\dot{W}_{\mathrm{comp}}} = \frac{P_{\mathrm{h}}}{P_{\mathrm{el}}}$ $\mathrm{COP}_{\mathrm{max}} = \frac{T_{\mathrm{h}}}{T_{\mathrm{h}} - T_{\mathrm{c}}}$
Standard enthalpy of formation	$\Delta h_{\mathrm{f}}^{0} = \Delta h_{\mathrm{r,formation\ from\ elements}}^{0}$
Standard free energy of formation	$\Delta g_{\mathrm{f}}^{0} = \Delta g_{\mathrm{r,formation\ from\ elements}}^{0}$
Third law entropy	$s^{0} = \int_{T=0\ \mathrm{K}}^{T=298.15\mathrm{K},} \frac{c_{\mathrm{p}}(T)}{T}\,\mathrm{d}T$
Standard enthalpy of reaction	$\Delta h_{\mathrm{r}}^{0} = \sum_{\mathrm{prod.}} \nu_{\mathrm{i}}\,\Delta h_{\mathrm{f,i}}^{0} - \sum_{\mathrm{reac.}} \nu_{\mathrm{j}}\,\Delta h_{\mathrm{f,j}}^{0}$
Standard free energy of reaction	$\Delta g_{\mathrm{r}}^{0} = \sum_{\mathrm{prod.}} \nu_{\mathrm{i}}\,\Delta g_{\mathrm{f,i}}^{0} - \sum_{\mathrm{react.}} \nu_{\mathrm{j}}\,\Delta g_{\mathrm{f,j}}^{0}$
Standard entropy of reaction	$\Delta s_{\mathrm{r}}^{0} = \sum_{\mathrm{prod.}} \nu_{\mathrm{i}}\,\Delta s_{\mathrm{i}}^{0} - \sum_{\mathrm{react.}} \nu_{\mathrm{j}}\,\Delta s_{\mathrm{j}}^{0}$

Electrochemistry

Name	Formula
Ohm's law	$\Delta E = RI$
Charge and current	$Q = It$
Power and current	$P = I\Delta E$
Energy (work) and charge	$W = Q\Delta E$
Energy (work), current and voltage	$W = I\Delta Et$
Faraday's law (of electrolysis)	The amount of mater converted is proportional to the charge converted: Moles converted: $n_{\text{con}} = \frac{Q}{nF}$ Mass converted: $m_{\text{con}} = \frac{MQ}{nF}$ Molar flow: $\dot{n}_{\text{con}} = \frac{I}{nF}$ Q = charge, n =no. of electrons, F = faraday's costant, M = molar mass, I = current.
Theoretical cell voltage	$E^0_{cell} = E^0_{electode\ 2} - E^0_{electde\ 1}$
Cell voltage	$E_{\text{rev,cell}}\, nF = -\Delta g_r$ $E_{\text{rev,cell}} = \frac{-\Delta g_r}{nF}$
Maximum work	$w_{max} = -\Delta g_r$
Maximum work	$w_{max} = qE_{rev,cell}$
Actual work	$w = qE_{actual,cell}$
Nernst equation (For negative Δg^0_r) ($E^0_{r,cell}$ assumed positive)	$E_{cell} = E^0_{r,cell} - \frac{RT}{nF}\ln\frac{\prod_{Prod.} a_i^{\nu_i}}{\prod_{React.} a_i^{\nu_i}}$
Nernst equation (For positive Δg^0_r) ($E^0_{r,cell}$ assumed positive)	$E_{cell} = E^0_{r,cell} + \frac{RT}{nF}\ln\frac{\prod_{Prod.} a_i^{\nu_i}}{\prod_{React.} a_i^{\nu_i}}$
Nernst, fuel cell or electrolyzer For equilibrium $H_2 + ½O_2 \rightleftarrows H_2O$ (approximated)	$E_{cell} = E^0_{cell} + \frac{RT}{2F}\ln p_{O2}^{½}\, p_{H2}$
Battery electrode capacity, single electrode (electrochemical equivalent)	$\text{ECE} = \frac{nF}{M}$ (C g^{-1}) $\text{ECE} = \frac{nF}{M \cdot 3600\ \text{s h}^{-1}}$ (Ah g^{-1})
Battery cell capacity, set of electrodes	$\frac{1}{\text{ECE}_{\text{cell.}}} = \frac{1}{\text{ECE}_{\text{anode.}}} + \frac{1}{\text{ECE}_{\text{cathode.}}}$

Solar Energy

Name	Formula
Solar "constant"	1361 W/m^2
Wavelength of an electromagnetic wave	$\lambda = \frac{c}{f}$ In which, • λ [m]: wavelength • c [m/s]: light speed (3x10^8 m/s) • f [Hz]: frequency
Energy of a photon	$E = h \cdot f$ In which, • E [J]: Energy of a photon • h [J/Hz]: Planck's constant (6.626x10^{-34} J/Hz) • f [Hz]: frequency
Electric power	$P = V \cdot I$ In which, • P [W]: electric power • V [V]: voltage • I [A]: electric current
Electric efficiency of a PV cell	$\eta = \frac{P}{P_{irradiance}}$ In which, • P [W]: electric power output • $P_{irradiance}$ [W]: solar irradiance power.
Fill factor	$FF = \frac{P}{V_{oc} \cdot I_{oc}}$ In which, • P [W]: electric power output • V_{oc} [V]: open circuit voltage • I_{sc} [A]: short circuit current
Shockley-Queisser limit	○ 33% for single-junction cell ○ 68% for multi-junction cells

Wind Energy

Name	Formula
Height effect on wind speeds	$$u(z) = u(z_1) \cdot \ln\left(\frac{z - z_1}{z_0}\right)$$ In which, • u(z) [m/s]: wind velocity at height z • $u(z_1)$ [m/s]: wind velocity at height z_1 • z [m]: height • z_1 [m]: reference height • z_0 [m]: roughness length
Weibull distribution	$$f(v) = \frac{k}{\lambda}\left(\frac{v}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{v}{\lambda}\right)^{k}\right)$$ In which, • f(v): probability of wind speed "v". • v [m/s]: wind speed • k: shape factor • λ: scale parameter
Mass flow rate of wind passing an area	$$\dot{m} = \rho A v$$ In which, • $\dot{m}$ [kg/s]: mass flow rate of air • ρ [kg/m^3]: mass density of air • A [m^2]: cross section area • v [m/s]: wind velocity
Total power from wind flow	$$P_{wind} = \frac{1}{2}\dot{m}v^2$$ In which, • $\dot{m}$ [kg/s]: mass flow rate of air • v [m/s]: wind velocity • P_{wind} [W]: total wind power
Wind power density	$$WPD = \frac{1}{2}\rho v^3$$ In which, • ρ [kg/m^3]: mass density of air • v [m/s]: wind velocity
Wind turbine power	$$P_t = C_p P_{wind}$$ In which, • P_t [W]: wind turbine power • C_p [-]: power coefficient • P_{wind} [W]: total wind power
Annual energy production (simplified)	$$AEP = P_{rated} \cdot t_{year} \cdot CF$$ In which, • AEP [W]: Annual energy production • P_{rated} [W]: Rated power of wind turbine • CF: Capacity factor

Angular velocity	$$\omega = \frac{v_{tip}}{R}$$ In which, • ω [rads/s]: angular velocity • v_{tip} [m/s]: blade tip speed velocity • R [m]: radius of rotation
Torque	$$T = F \cdot R$$ In which, • F [N]: Linear force • R [m]: radius of rotation • T [N.m]: Torque
Power	$$P = T \cdot \omega$$ In which, • P [W]: power • T [N.m]: torque • ω [rads/s]: angular velocity
Tip speed ratio	$$\lambda_{TSR} = \frac{\omega R}{v}$$ In which, • ω [rads/s]: angular velocity • R [m]: radius of rotation • v [m/s]: wind speed

Bioenergy

Name	Formula
Relationship between heating values	$$HHV = LHV + h_{H_2O,fg}\left(\frac{n_{H_2O,out}}{n_{fuel,in}}\right)$$ In which, • HHV [J/mol]: Higher heating value • LHV [J/mol]: Lower heating value • $h_{H_2O,fg}$ [J/mol]: Enthalpy of vaporization • $n_{H_2O,out}$ [mol]: water vaporized in the combustion products. • $n_{fuel,in}$ [mol]: fuel input in the combustion.
Lower heating value in humid basis	$$LHV_{humid} = (1 - x_{H_2O})LHV_{dry} - x_{H_2O}h_{fg}$$ In which, • LHV_{humid} [J/mol]: Lower heating value on humid basis • x_{H_2O}: molar fraction of water • $h_{H_2O,fg}$ [J/mol]: Enthalpy of vaporization
Air fuel ratio	$$\phi = \frac{\left(\frac{n_{fuel}}{n_{air}}\right)_{actual}}{\left(\frac{n_{fuel}}{n_{air}}\right)_{st}}$$ In which, • ϕ [-]: Air-fuel ratio • $\left(\frac{n_{fuel}}{n_{air}}\right)_{actual}$: actual molar ratio between fuel and air • $\left(\frac{n_{fuel}}{n_{air}}\right)_{st}$: stoichiometric ratio between fuel and air for complete combustion
Buswell equation	$$C_cH_hO_o + \left(c - \frac{h}{4} - \frac{o}{2}\right)H_2O \rightarrow \left(\frac{c}{2} + \frac{h}{8} - \frac{o}{4}\right)CH_4 + \left(\frac{c}{2} - \frac{h}{8} + \frac{o}{4}\right)CO_2$$ In which, • c: number of carbon atoms per mole of organic matter • h: number of hydrogen atoms per mole of organic matter • o: number of oxygen atoms per mole of organic matter

Nuclear physics

Name	Formula
Einstein's mass and energy relation	$$E = mc^2$$
Isotopes $^{y}_{z}X$	X: element symbol, y: nucleons, z: protons
Mass defect	$$\Delta m = \sum_{protons} m + \sum_{neutrons} m + \sum_{electrons} m - \text{actual atomic mass}$$ Instead of protons and electrons, the mass of the same number of hydrogen atomic masses (protium, $^{1}_{1}H$) can be used

Infrastructure

Name	Formula
Root mean square voltage	$$\bar{V}_{rms} = \frac{V_{peak}}{\sqrt{2}}$$ In which, • $\bar{V}_{rms}$ [V]: root mean square voltage • V_{peak} [V]: peak amplitude of voltage.
Root mean square current	$$\bar{I}_{rms} = \frac{I_{peak}}{\sqrt{2}}$$ In which, • $\bar{I}_{rms}$ [V]: root mean square current • I_{peak} [V]: peak amplitude of current.
Ideal transformer	$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$ In which, • V_p [V]: Voltage in the primary cell • V_s [V]: Voltage in the secondary cell • N_p [V]: Number of turns in the primary cell • N_s [V]: Number of turns in the secondary cell
Power loss due to ohmic heating	For DC: $$P_{loss} = R_{cable} \cdot I^2$$ For AC: $$P_{loss} = R_{cable} \cdot I_{rms}^2$$ In which, • P_{loss} [W]: energy loss due to ohmic heating • R_{cable} [W]: equivalent ohmic resistance of the transmission • I [A]: current • $\bar{I}_{rms}$ [A]: root mean square of current.
Apparent power	$$P_{app} = V_{rms} I_{rms}$$ In which, • $\bar{V}_{rms}$ [V]: root mean square voltage • $\bar{I}_{rms}$ [A]: root mean square of current. • P_{app} [W]: apparent power
Power factor (for linear systems, single phase)	$$PF = \frac{P_{real}}{P_{app}} = \cos(\theta)$$ In which, • P_{app} [W]: apparent power • P_{real} [W]: real power • θ [deg]: phase difference between voltage and current • PF: power factor.

Economics

Name	Formula
Present value of future single payment	$$P = \frac{F}{1+i}$$ In which, • P [$]: present value • F [$]: future value • i : discount rate
Discount cash flow	$$DCF = \frac{CF}{(1+i)^n}$$ In which, • DCF [$]: discounted cash flow • CF [$]: cash flow • i : discount rate • n: year that the cash flow takes place
Weighted average cost of capital	$$i_{WACC} = \gamma_d i_d + \frac{1-\gamma_d}{1-\tau} i_e$$ In which, • i_{WACC}: weighted average cost of capital • γ_d: debt fraction • i_d: debt discount rate • τ: corporate tax • i_e: equity discount rate
Effective discount rate	$$i^{eff} = \frac{i-f}{i+f}$$ In which, • i^{eff}: effective discount rate • i: discount rate • f: inflation rate
Conversion of single present payment into annuity (capital recovery factor)	$$f_{A/P} = \frac{i(i+i)^n}{(1+i)^n - 1}$$ In which, • i: discount rate • $f_{A/P}$ [1/y]: conversion factor • n: number of years of amortization
Converting an annuity into a single present payment	$$f_{P/A} = \frac{1}{f_{A/P}}$$ In which, • $f_{A/P}$ [1/y]: conversion factor from present payment to annuity • $f_{P/A}$ [y]: conversion factor from annuity to present payment

Economy of scale	$$C = C_0 \left(\frac{S}{S_0}\right)^{\alpha}$$ In which, • C [$]: Equipment cost • C_0 [$]: Equipment reference cost • S [unit*]: Scale of the equipment • S_0 [unit*]: Reference scale of the reference cost • α: economy of scale factor *"unit refers to a certain reference unit, such as kWh or area, from which the scale is measured.
Learning curve	$$C = C_0 \cdot X^{-\log_2(b)}$$ In which, • b: progress ratio • C [$]: Equipment cost • C_0 [$]: Equipment reference cost • X [unit*]: cumulative installed capacity *"unit refers to a certain reference unit, such as kWh or area, from which the installed capacity is measured.
Levelized cost – general definition	$$c = \sum_n \frac{CF_n}{(1+i)^n} / \sum_n \frac{R_n}{(1+i)^n}$$ In which, • c [$/unit*] : levelized • CF_n [$]: cash flow at year n • i : discount rate • n : year • R_n [unit*]: total production in year n *"unit refers to a certain reference unit, such as kWh or kg, from which the cost is measured.

Levelized cost – simplified calculation	$$c = \frac{CAPEX \cdot f_{A/P} + OPEX_{fix}}{R_p \cdot t_{op,year}} + \sum\left(\frac{R_i}{R_p}\right) c_i - \sum\left(\frac{R_{bp}}{R_p}\right) c_{bp}$$ In which, • c [$/unit*]: levelized • CAPEX [$]: capital expenses • $f_{A/P}$ [1/y]: conversion factor from present payment to annuity • $OPEX_{fix}$ [$/y]: annual fixed operational expenses • R_p [unit*/h]: hourly rate production • $t_{op,year}$[h/y]: full load equivalent operating hours in a year • R_i [unit*/h]: hourly rate of consumption of input "i" • R_{bp} [unit*/h]: hourly rate of consumption of by-product "bp" • c_i [$/unit*]: specific cost of input "i". • c_{bp} [$/unit*]: specific cost of by-product "bp". *"unit refers to a certain reference unit, such as kWh or kg, from which the cost is measured.
Net present value – general definition	$$NPV = \sum_n \frac{CF_n}{(1+i)^n}$$ In which, • CF_n [$]: cash flow at year n • i : discount rate • n : year
Internal rate of return	$$NPV = \sum_n \frac{CF_n}{(1+irr)^n} = 0$$ In which, • CF_n [$]: cash flow at year n • irr : internal rate of return • n : year